

Unit 20: Concavity and Curve Sketching

1. The idea, in plain words

You already know how to tell whether a graph is climbing or falling using $f'(x)$. This unit adds one more layer of detail: is the graph curving upward like a cup (concave up), or curving downward like a frown (concave down)? That extra layer of shape information comes from the second derivative, $f''(x)$ — the derivative of the derivative.

Picture it like this: a cup holds water — it curves upward, bending away from a straight line underneath it. A frown-shape spills water — it curves downward, bending away from a straight line above it. Concave up ($f''(x) > 0$) looks like a cup; concave down ($f''(x) < 0$) looks like a frown.

Where the concavity switches from one to the other, that's called an inflection point. Picture an S-curve: it starts curving one way, then smoothly flips to curving the other way — the exact spot where it flips is the inflection point.

A bonus tool this unit gives you: the Second Derivative Test. Instead of always doing a full sign chart with the first derivative to classify a critical point, you can sometimes just check the sign of f'' right at that point: a cup-shape ($f'' > 0$) sitting at a critical point means you're at the bottom of a dip — a local min. A frown-shape ($f'' < 0$) at a critical point means you're at the top of a hill — a local max. It's often much faster than a full sign chart — but it has a blind spot: if $f''(c) = 0$ exactly, the test tells you nothing, and you have to fall back on the first derivative test instead.

Putting it all together — full curve sketching. By now you have every tool needed to build a complete, accurate picture of any function's graph without ever plotting a single point by brute force:

- Domain — where is the function even defined?
- Asymptotes — horizontal and vertical (from Unit 6).
- $f'(x)$ — where is it increasing/decreasing, and where are the local max/min points?
- $f''(x)$ — where is it concave up/down, and where are the inflection points?

Combine all four, and you can describe (or sketch) the entire shape of the graph with total confidence.

2. Toolbox

Concavity Test:

$$f''(x) > 0 \text{ on an interval} \implies f \text{ is concave up there (cup shape)}$$

$$f''(x) < 0 \text{ on an interval} \implies f \text{ is concave down there (frown shape)}$$

Inflection point: a point where the concavity actually changes — found where $f''(x) = 0$ or $f''(x)$ is undefined, and the sign of f'' genuinely switches on either side (just like critical points, finding a candidate isn't enough — you must confirm the sign actually flips).

Second Derivative Test (a shortcut for classifying a critical point c where $f'(c) = 0$):

- If $f''(c) > 0$: local minimum at c (cup shape — the bottom of a dip).
- If $f''(c) < 0$: local maximum at c (frown shape — the top of a hill).
- If $f''(c) = 0$: the test is inconclusive — fall back on the First Derivative Test instead.

The complete curve-sketching checklist:

1. Domain of f .
2. Asymptotes: horizontal (compare degrees / take limits at $\pm\infty$) and vertical (denominator zero without cancellation).
3. $f'(x)$: find critical points, build a sign chart, identify increasing/decreasing intervals and local max/min points.
4. $f''(x)$: find candidate inflection points, build a sign chart, identify concave up/down intervals and genuine inflection points.
5. Combine everything into a full description (or sketch) of the graph.

3. Common mistakes

- Mixing up which sign of f'' means which shape. $f'' > 0$ is concave up (cup); $f'' < 0$ is concave down (frown). It's easy to flip these under pressure — the “cup holds water” picture is the fastest way to double-check yourself.
- Calling every point where $f'' = 0$ an inflection point, without checking for an actual sign change. Just like with critical points and extrema, a candidate point isn't automatically the real thing — you must verify concavity genuinely flips on either side.
- Using the Second Derivative Test when $f''(c) = 0$. This case gives you no information at all — you have to go back to the First Derivative Test for that specific point.
- Confusing concavity information with increasing/decreasing information. These come from two different derivatives (f' for direction, f'' for curviness) — a function can be increasing and concave down at the same time, or decreasing and concave up, and so on. All four combinations are possible.
- Forgetting to check for asymptotes on a rational function before finishing a full sketch — a curve-sketching problem involving a fraction almost always has at least a horizontal asymptote to report.

4. Problem Set

Warm-up

For each function, find the intervals of concavity and any inflection points.

1. $f(x) = x^3 - 3x^2 + 2$
2. $f(x) = x^4 - 6x^2$
3. $f(x) = x^3 - 6x^2 + 9x$
4. $f(x) = -x^3 + 3x$
5. $f(x) = x^4 - 4x^3 + 1$
6. $f(x) = x^3 - 3x + 1$

Standard

7. $f(x) = x^3 - 6x^2 + 9x + 1$ — use the Second Derivative Test to classify each critical point.
8. $f(x) = x^4 - 4x^3 + 2$ — use the Second Derivative Test where possible, and note where it's inconclusive.
9. $f(x) = 2x^3 - 3x^2 - 12x + 5$ — find the inflection point and concavity intervals.
10. $f(x) = x^4 - 2x^2 + 3$ — find concavity intervals and inflection points.
11. $f(x) = x^3 - 3x^2 - 9x + 1$ — use the Second Derivative Test to classify each critical point.

12. $f(x) = 3x^5 - 5x^3$ — find concavity intervals and all inflection points.

Challenge

For Problems 13–17, give a full analysis: domain, asymptotes (if any), increasing/decreasing intervals, local extrema, concavity intervals, and inflection points.

13. $f(x) = -x^3 + 6x^2 - 9x + 3$

14. $f(x) = \frac{(x+1)^2}{1+x^2}$

15. $f(x) = x^4 - 4x^3 + 4x^2$

16. $f(x) = (x-2)^3$

17. $f(x) = \frac{1}{x^2 + 1}$

Applied

18. A company's cost (in dollars) is $C(x) = x^3 - 9x^2 + 30x + 50$. Find the inflection point of the cost curve, and interpret it as the production level where the cost's growth transitions from "increasing at a decreasing rate" (diminishing marginal cost growth) to "increasing at an increasing rate" (accelerating marginal cost growth).
19. A new product's number of users is modeled by $N(t) = -t^3 + 15t^2 + 100$ for t in $[0, 10]$ months. Find the inflection point, and interpret it as the moment of fastest growth, after which growth continues but begins to slow.
20. An object's position is $s(t) = t^3 - 6t^2 + 9t$. Find the inflection point of the position curve, and explain what it tells you about the object's acceleration at that moment (connect this back to the "speeding up/slowing down" ideas from Unit 11).
21. A company's profit is $P(x) = -x^3 + 12x^2 - 20x$. Find the inflection point, and interpret it as the production level where returns begin to diminish (marginal profit switches from increasing to decreasing).